

# Shakti Banik

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## EDUCATION

**Master of Science in Mechanical Engineering** August 2025 – May 2027 (Expected)  
*Washington State University – Vancouver* Vancouver, WA

- CGPA – 4.00/4.00

**Bachelor of Science in Mechanical Engineering** March 2018 – May 2023  
*Bangladesh University of Engineering and Technology (BUET)* Dhaka, Bangladesh

- CGPA – 3.52/4.00 (81<sup>st</sup> out of 189 students)
- University Merit Scholarship for academic excellence in one undergraduate semester.

## RESEARCH INTERESTS

|                     |                            |                             |
|---------------------|----------------------------|-----------------------------|
| Micro/Nanosystems   | Biomedical Devices         | Biosensors & Bioelectronics |
| Wearable Biosensors | Materials Characterization | Biomedical Imaging          |

## RESEARCH EXPERIENCE

**Graduate Research** August 2025 – Present

**Title:** Development of a Paper-Based Colorimetric Biosensor for Cannabidiol (CBD) Monitoring Toward Pain Management Applications

**Supervisor:** Dr. Jong-Hoon Kim, Associate Professor, Department of Mechanical Engineering, WSU Vancouver

- **Developed and experimentally evaluated** a low-cost paper-based colorimetric biosensor for CBD detection using Fast Blue B chemistry and smartphone imaging.
- **Developed a CIELAB-based digital image analysis method** for blank-referenced color segmentation and signal quantification.
- **Evaluated sensor performance** in CBD reference solutions and CBD-spiked saliva, with detectable responses down to 0.05  $\mu\text{g/mL}$  and 0.1  $\mu\text{g/mL}$ , respectively.
- **Preparing for human-subject testing** with CBD-treated patients to evaluate sensor performance using clinical saliva samples.

**Undergraduate Research** May 2022 – May 2023

**Title:** Design and Analysis of a Bladeless Air Jet Windscreen Wiper System

**Supervisor:** Dr. Md. Ehsan, Professor, Department of Mechanical Engineering, BUET

- **Designed and experimentally evaluated** a bladeless air-jet windscreen wiper system using different nozzle geometries.
- **Modeled rain and airflow characteristics** using experimental measurements and MATLAB, and assessed visibility and power consumption.

## PUBLICATIONS

- **Banik, S., Fales, J., & Kim, J.-H.** (2026). Biosensing technologies for cannabinoid monitoring in pain management: Current methods, clinical needs, and translational challenges. *Journal of Pharmaceutical and Biomedical Analysis*, 281, 117673. <https://doi.org/10.1016/j.jpba.2026.117673>

## CONFERENCE ABSTRACTS

- **Banik, S.** and Kim, J.-H., “Development of a Paper-Based Colorimetric Biosensor for Cannabidiol Monitoring Toward Pain-Management Applications,” *ASME 2026 International Mechanical Engineering Congress and Exposition (IMECE 2026)*, Vancouver, British Columbia, Canada, November 8–12, 2026. (Accepted)

## TEACHING & PROFESSIONAL EXPERIENCE

**Graduate Teaching Assistant** August 2025 – Present  
*Washington State University Vancouver, School of Engineering and Computer Science* Vancouver, WA

- **Teaching Assistant for five undergraduate engineering courses:** MECH 211 Statics, MECH 215 Mechanics of Materials, MECH 304 Introduction to Electronic Circuits, MECH 310 Manufacturing Processes, and MECH 309 Engineering Materials.
- **Serve as laboratory instructor for MECH 309 Engineering Materials**, leading hands-on experiments in dimensional metrology, composite characterization, optical microscopy, and Instron mechanical testing.
- **Led two hands-on LTspice demonstration sessions** for MECH 304 Introduction to Electronic Circuits, teaching students circuit simulation and basic software use.
- **Support undergraduate instruction** through grading, office hours, laboratory supervision, technical guidance, and individualized assistance with engineering concepts and data analysis.

#### Industrial Trainee

October 2022 – November 2022

*Synovia Pharma PLC (formerly Sanofi Bangladesh Limited)*

*Dhaka, Bangladesh*

- **Analyzed water losses** in a pharmaceutical effluent treatment plant (ETP) and recommended calibration of process measurement instruments.
- **Developed a standardized condition-based monitoring profile** and supported troubleshooting of industrial HVAC systems.

#### PROJECTS

##### Arogya: Smart Medicine Dispensing Box for Elderly Care

Apr 2021 – Aug 2021

- Developed an Arduino-based medicine dispenser with automated alerts and Bluetooth-controlled scheduling through a caregiver-integrated Android app.

#### TECHNICAL SKILLS

- **Experimental & Biosensing:** Experimental design, biosensor development, biological sample testing.
- **Image & Data Analysis:** Image processing, statistical analysis, MATLAB, Python.
- **Cleanroom & Microfabrication:** Photolithography, spin coating, wet chemical etching, RF magnetron sputtering, thin-film processing.
- **Characterization & Instrumentation:** Optical microscopy, ellipsometry, optical/stylus profilometry, Instron mechanical testing.
- **Research Compliance:** CITI Human Subjects Research (Biomedical Research); Biomedical Responsible Conduct of Research (RCR).

#### TEST SCORE

**IELTS Academic:** Overall 8.0 (Listening: 8.0, Reading: 8.0, Writing: 8.0, Speaking: 7.0) October 2024

#### RELEVANT COURSEWORK

Design of Experiments

MEMS Engineering

Lab-on-a-Chip

Machine Learning and AI for Engineers

#### SELECTED AWARDS & LEADERSHIP

- **Vice President, BUET Photographic Society** May 2022 – June 2023
- **Tareque Masud Best Emerging Director Award**, 15th International Inter-University Short Film Festival November 2023
- **Best Project Award**, Dhaka DocLab September 2023
- **Academic Excellence Award**, University Merit January 2018

#### REFERENCES

**Dr. Jong-Hoon Kim**

Associate Professor, Mechanical Engineering

Washington State University Vancouver

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**Dr. Praveen Sekhar**

Associate Professor, Electrical Engineering

Washington State University Vancouver

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